

H.I.T COLLEGE OF ENGINEERING

Academic College Management System (CMIS)

Complete Project Documentation & System Manual

Author: Development Team

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1. System Overview

The H.I.T College of Engineering Academic College Management System (CMIS) is a unified, role-based administrative portal and student portal designed to manage operations across all departments of the institution. Built using the modern MERN stack (MongoDB, Express, React, Node.js), it automates critical operations including student enrollment, attendance management, marks tracking, fee calculation, salary processing, and lab infrastructure cataloging.

Key Architecture & Tech Stack

- Frontend: React.js with Vanilla CSS for custom, high-fidelity user interfaces.
 - Backend: Node.js with Express.js supporting modular, controller-based routing structure.
 - Real-Time Messaging: Socket.io powering persistent chat services between staff and students.
 - Database: MongoDB utilizing Mongoose schemas to model schema constraints.
 - Cloud Integration: Cloudinary API integration for student and faculty file/avatar uploads.
 - API Specs: Swagger JSDoc UI documentation for developers at </api/v1/docs>.
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2. Core Modules & System Features

2.1 Authentication & Security

The platform features a secure authentication mechanism supporting dual paths: traditional JWT-based login for standard students/admins, and a secure OTP-based login process for staff. Password strength validation and registration format checks ensure data cleanliness on onboarding.

2.2 Smart Fees Calculation Engine

A department-specific tuition fees engine regulates and calculates student balance amounts:

Department Type	Tuition Fees Threshold
• Computing Departments (CSE, CSBS, etc.)	Fees >= INR 50,000
• Non-Computing Departments (ECE, Mech, etc.)	Fees: INR 30,000

2.3 Marks Ledger & Attendance Management

Administrators and Staff can process marks and attendance grades. Students can only view their marks ledger once they have successfully attended tests tracked in the LMS. The admin interface fetches dynamic, role-restricted student profiles to ensure staff/admin data parity.

2.4 Salary & Payroll Components

Payroll processing tracks fixed salaries, allowances, and bonuses for HODs, Faculty Members, Office Staff, and the Principal. Built-in controllers enforce validation checks and log transaction histories for auditing purposes.

3. Laboratory Infrastructure Catalog

The Lab Infrastructure directory tracks details of systems, maintenance history, and department assignments. Staff and HODs can monitor equipment counts and check statuses in real time. Static datasets stored in the systems include:

Lab Code	Lab Name	Dept	Capacity	Systems	In-Charge
CSE101	Programming Laboratory	CSE	60	60	Dr. R. Kumar
CSE102	Database Management Systems Lab	CSE	60	60	Prof. S. Priya
CSBS101	Business Analytics Lab	CSBS	50	50	Dr. K. Saravanan

4. API Routing & Client Endpoints

4.1 Server Routes (API v1)

- POST /api/v1/auth/register - Onboards new staff & students.
- POST /api/v1/auth/login - Handles credential-based JWT issuance.
- GET /api/v1/office/faculty/:id - Returns details for specific faculty.
- GET /api/v1/labs - Catalog listing with departmental filters.
- GET /api/v1/attendance - Retrieves class logs for validation.

4.2 Client Pages

- **/login** - Unified landing gateway featuring an Accordion component.
- **/cmis/marks** - Dynamic ledger enabling marks registration and lookups.
- **/cmis/labs** - Lab infrastructure manager with statuses and inventory tables.
- **/cmis/office-salary** - Process faculty salary transactions securely.

